

# CI/CD 101

with Gitlab 

30/04/2025

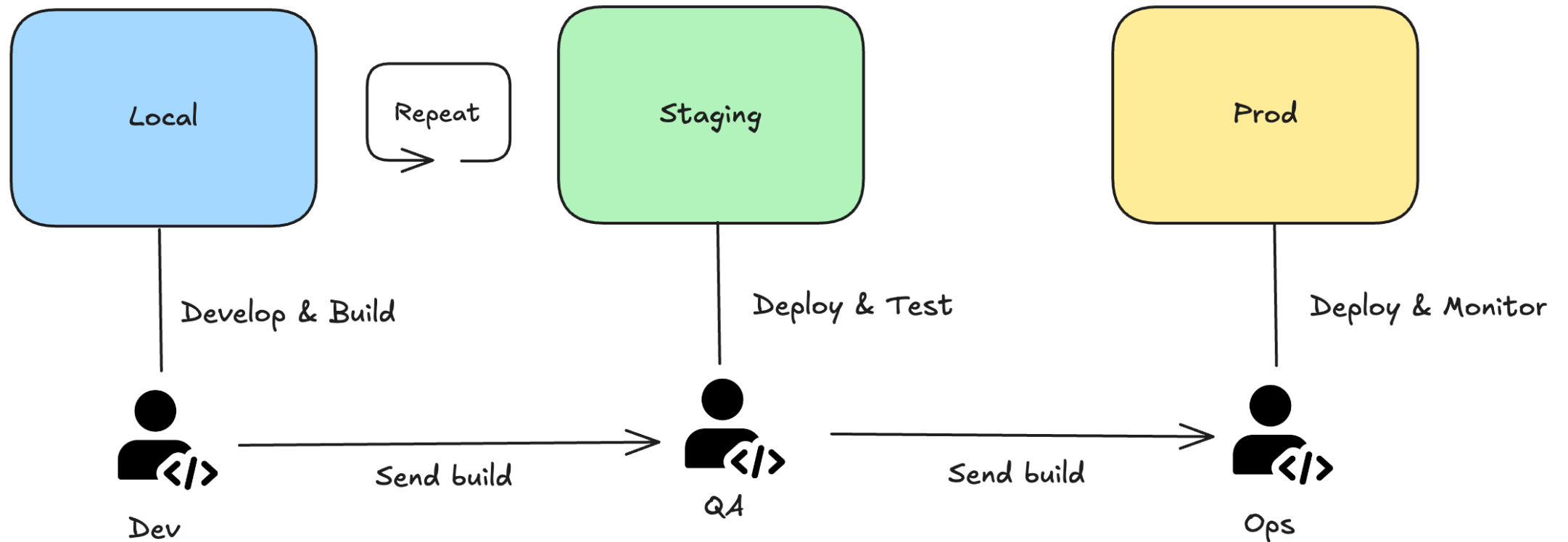
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## 01. What is CI/CD ?

# Life without CI/CD

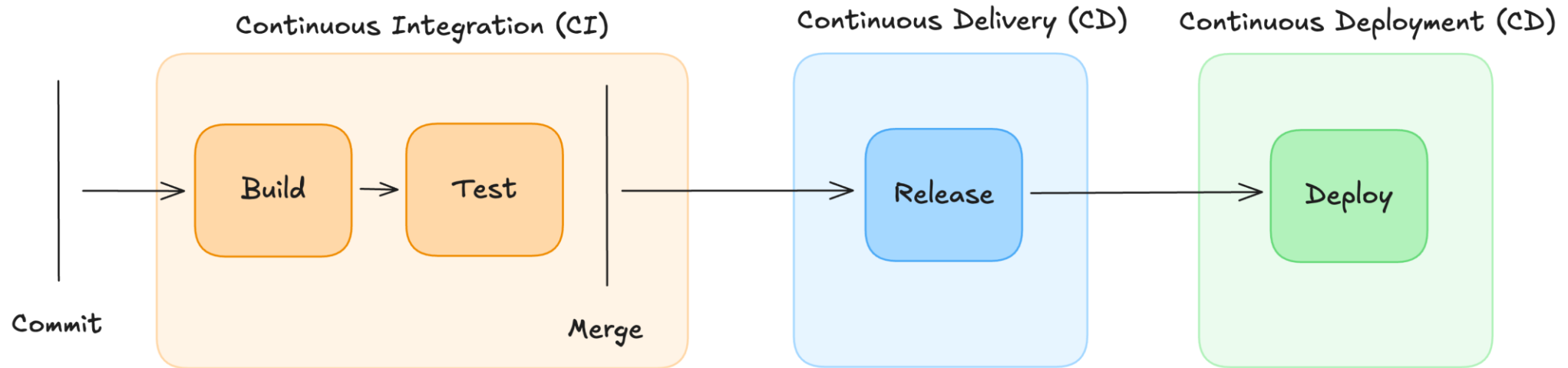
01. What is CI/CD ?



# Definition

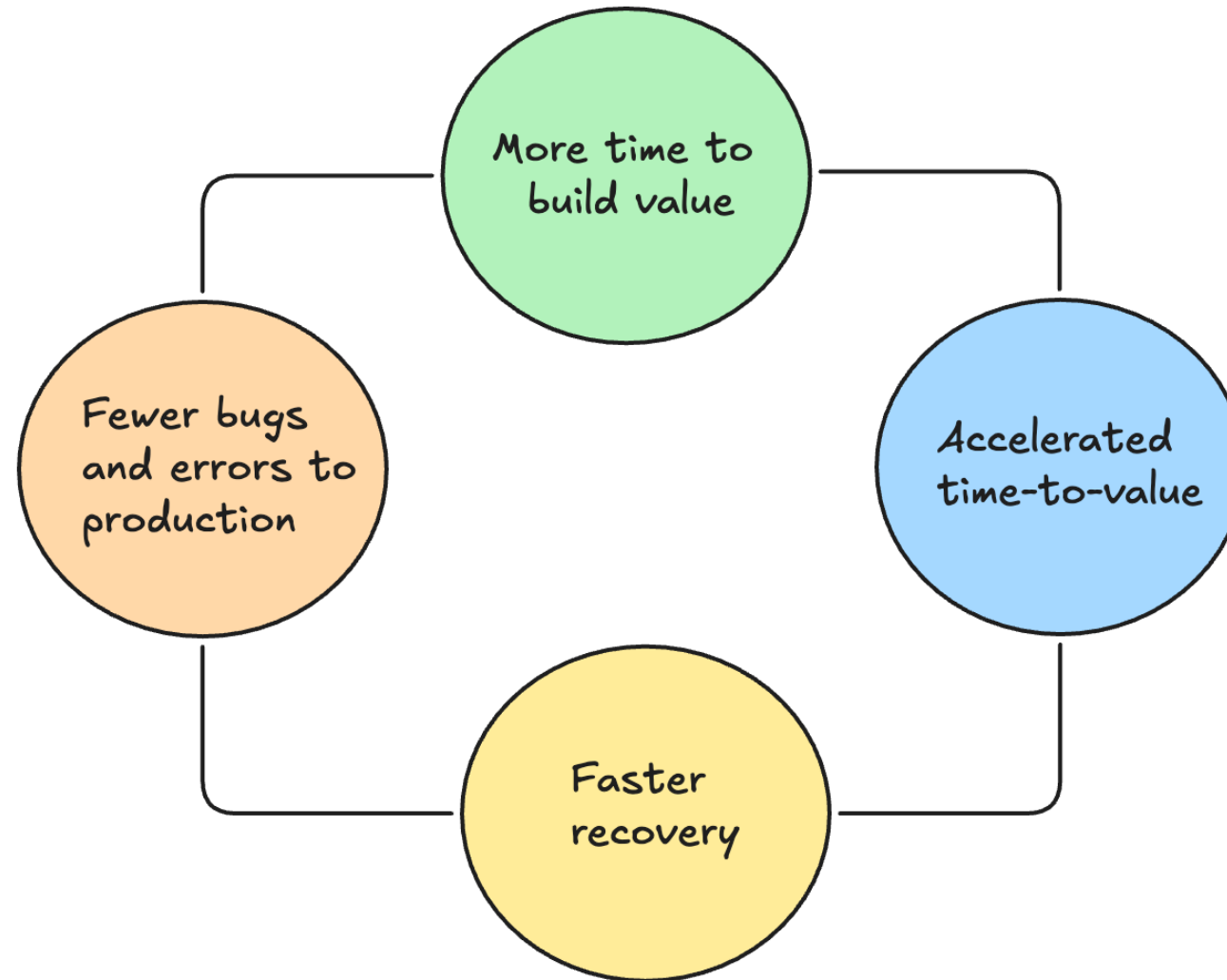
01. What is CI/CD ?

«CI/CD automates much or all of the manual human intervention traditionally needed to get new code from a commit into production» [Gitlab](#)



# Life with CI/CD

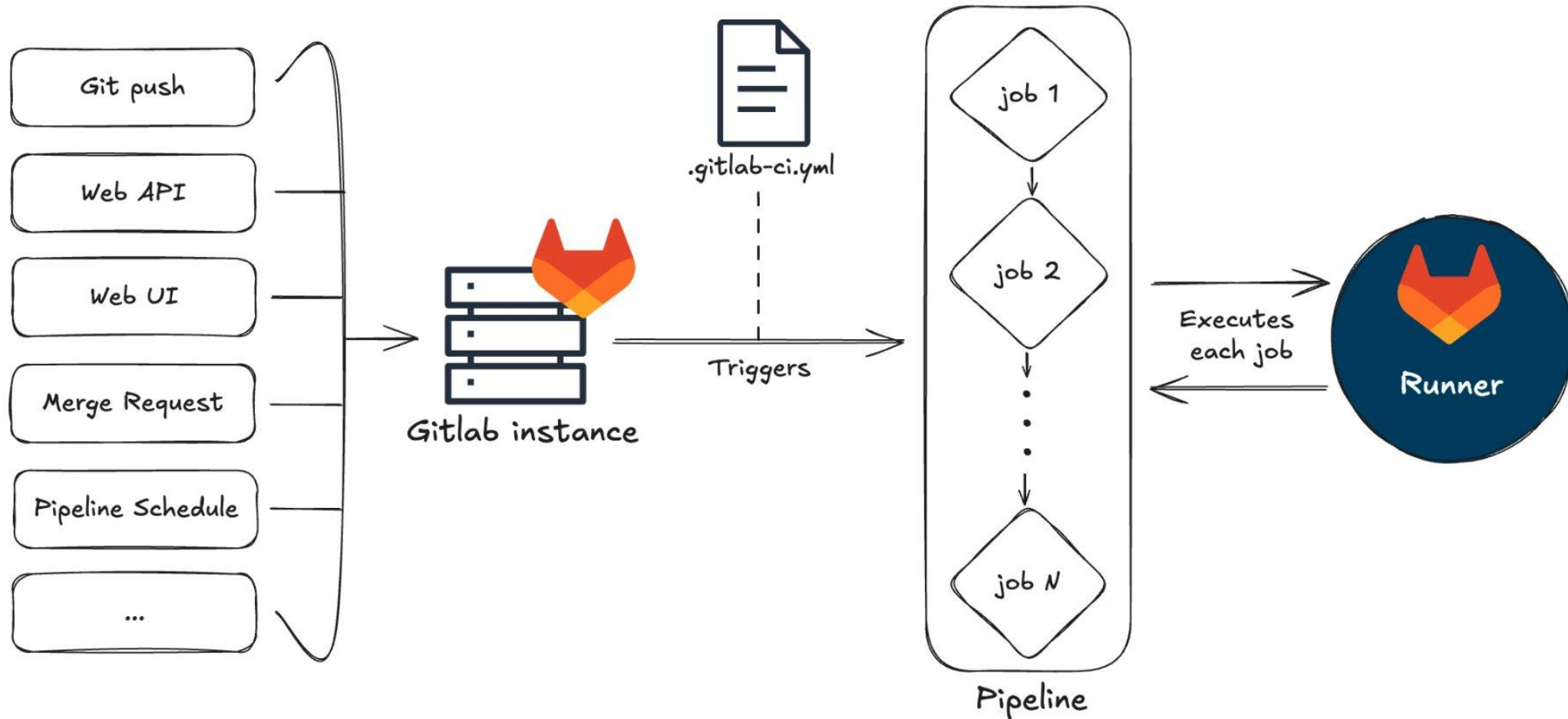
01. What is CI/CD ?



## 02. Introduction to Gitlab CI/CD

# Concepts

## 02. Introduction to Gitlab CI/CD





# Configuration

The .gitlab-ci.yml file

02. Introduction to Gitlab CI/CD



Stages →

Job →

```
1 variables:
2   APP_NAME: "webapp"
3
4 stages:
5   - build
6   - test
7
8 build:
9   stage: build
10  script:
11    - echo "Building the application ${APP_NAME}"
12
13 test:
14   stage: test
15   script:
16     - echo "Running tests ..."
17
18
```

← Variables

← Script

# Configuration

## Sequencing

## 02. Introduction to Gitlab CI/CD

 .gitlab-ci.yml

```
1 variables:
2   APP_NAME: "webapp"
3
4 stages:
5   - build
6   - test
7   - deploy
8
9 build:
10  stage: build
11  script:
12    - echo "Building the application ${APP_NAME}"
13    - mkdir build_output
14    - echo "Compiled app" > build_output/app.txt
15  artifacts:
16    paths:
17      - build_output/
18
19 test:unit:
20  stage: test
21  script:
22    - echo "Running tests ..."
23    - cat build_output/app.txt
24
25 test:func:
26  stage: test
27  script:
28    - echo "Running tests ..."
29  needs: [test:unit]
30
31 deploy:
32  stage: deploy
33  script:
34    - echo "Deploying the application ${APP_NAME}"
35  dependencies:
36    - build
37
```

Artifacts →

Dependencies →

← Needs

# Configuration

## Conditioning

## 02. Introduction to Gitlab CI/CD

 .gitlab-ci.yml

```
1 variables:
2   APP_NAME: "webapp"
3
4 stages:
5   - build
6   - test
7   - deploy
8
9 build:
10  stage: build
11  script:
12    - echo "Building the application ${APP_NAME}"
13    - mkdir build_output
14    - echo "Compiled app" > build_output/app.txt
15  artifacts:
16    paths:
17      - build_output/
18
19 test:
20  stage: test
21  script:
22    - echo "Running tests ..."
23    - cat build_output/app.txt
24  rules:
25    - changes:
26      - "src/x*/*.js"
27      - "tests/**/*"
28  dependencies:
29    - build
30
31 deploy:
32  stage: deploy
33  script:
34    - echo "Deploying the application ${APP_NAME}"
35  rules:
36    - if: '$CI_COMMIT_BRANCH = "master"'
37      when: manual
38    - if: '$CI_COMMIT_BRANCH = "staging"'
39  needs:
40    - job: test
41
```

Changes →

Rules

If →

When

# Configuration

## Templating

## 02. Introduction to Gitlab CI/CD

Include

 .gitlab-ci/templates/build.yml

```
1 .build_template:
2   script:
3     - echo "Building the application ${APP_NAME}"
4     - mkdir build_output
5     - echo "Compiled app" > build_output/app.txt
```

 .gitlab-ci.yml

```
1 include:
2   - local: '.gitlab-ci/templates/build.yml'
3
4 variables:
5   APP_NAME: "webapp"
6
7 stages:
8   - build
9   - test
10  - deploy
11
12 build:
13   stage: build
14   extends: .build_template
15   artifacts:
16     paths:
17       - build_output/
18
19 test:
20   stage: test
21   script:
22     - echo "Running tests ..."
23     - cat build_output/app.txt
24   rules:
25     - changes:
26       - "src/**/*.js"
27       - "tests/**/*.js"
28   dependencies:
29     - build
30
31 deploy:
32   stage: deploy
33   script:
34     - echo "Deploying the application ${APP_NAME}"
35   rules:
36     - if: '$CI_COMMIT_BRANCH = "master"'
37       when: manual
38     - if: '$CI_COMMIT_BRANCH = "staging"'
39   needs:
40     - job: test
41
```

Extends

# Contact



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